

1 MSDS: Materials Safety Data Sheets

1. standard documents that provide safety info on Materials
2. provided by suppliers
3. include info on:
 - composition
 - handling + PPE (Personal protective equipment)
 - hazards
 - shipping regulations

2 Legislation

- can support all 3 capitals
- includes state, national, and international standards+agreements

3 EV case study

People:

- range anxiety
-

Planet:

- Concerns over the carbon emission of energy grid
- are carbon emissions better for EVs?
- legislation in EU requires battery recycling

Prosperity:

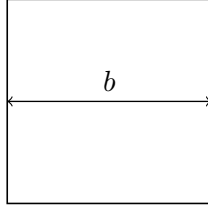
- Li only produced by 2 major countries
- current Li production does not meet anticipated demand
- Nd production too low
- 95% of Nd from one country(China)

4 Final presentation material index

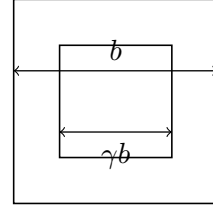
Objective - Minimize deflection δ :

$$\delta = \frac{F_f L^3}{C_1 EI}$$

Since F_f , L , and C_1 are constants, $\delta \propto EI$



A_s (solid square)



A_h (hollow square)

For the solid square section (A_s) with side b :

$$I_s = \frac{b^4}{12}, \quad \text{and since } A_s = b^2, \quad I_s = \frac{A_s^2}{12}.$$

For the hollow square section (A_h) with outer side b and inner side γb :

$$I_h = \frac{b^4 - \gamma^4 b^4}{12} = \frac{(1 - \gamma^4) b^4}{12}.$$

Using the solid reference area $A_s = b^2$,

$$I_h = \frac{(1 - \gamma^4) A_s^2}{12}.$$

Introduce the bending shape factor ϕ (normalized so a solid square gives $\phi = 1$):

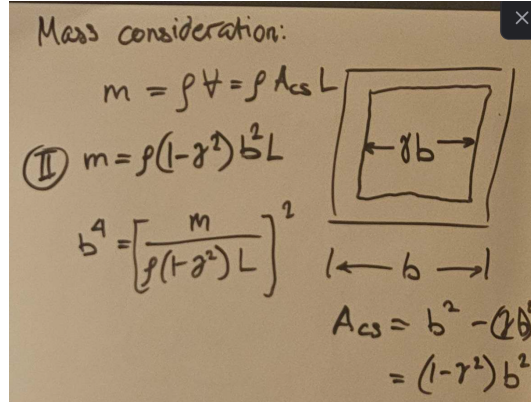
$$\phi = \frac{12 I}{A_s^2} \quad \Rightarrow \quad \begin{cases} \phi_s = \frac{12 I_s}{A_s^2} = 1, \\ \phi_h = \frac{12 I_h}{A_s^2} = 1 - \gamma^4. \end{cases}$$

Thus,

$$\boxed{\phi = \frac{12 I}{A_s^2}}$$

encapsulates bending efficiency; it equals 1 for the solid square and decreases with hollowness for fixed reference area A_s .

Mass as a constraint:



From $m = \rho V$:

$$b^4 = \left[\frac{m}{\rho(1-\gamma^2)L} \right]^2$$

expanding Inertia term:

$$I = \frac{b^4}{12} - \frac{\gamma^4 b^4}{12}$$

$$I = \frac{(1-\gamma^4)}{12} b^4$$

combining inertia and mass equations:

$$I = \frac{(1-\gamma^4)}{12(1-\gamma^2)^2} \frac{m^2}{\rho^2 L^2}$$

Plug inertia term back into objective function:

$$\delta = \frac{F_f L^3}{C_1 E} \frac{12(1-\gamma^2)^2}{(1-\gamma^4)} \frac{\rho^2 L^2}{m^2} = \frac{F_f L^3}{C_1} \frac{12(1-\gamma^2)^2}{(1-\gamma^4)} \frac{L^2}{m^2} \left(\frac{\rho^2}{E} \right)$$

material index M :

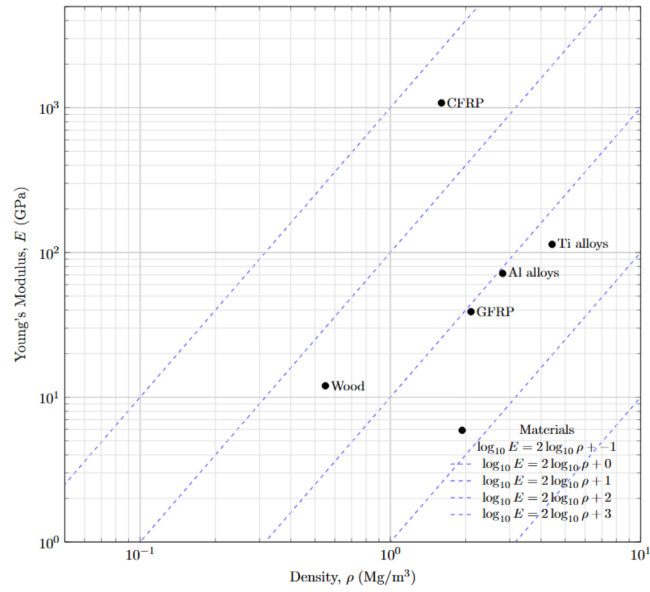
$$M = \frac{\rho^2}{E}$$

$$M_h = \frac{(1 - \gamma^2)^2}{(1 - \gamma^4)} \frac{\rho^2}{E}$$

$$M_h = \phi \frac{\rho^2}{E}$$

$$\log M = 2 \log \rho - \log E$$

$$\log E = 2 \log \rho - \log M$$



<https://hal.science/jpa-00251707/document>